

DIGITAL TWIN COMMAND CENTER

Dashboard 3 — Multi-Echelon Simulation • Inventory Dynamics • What-if Scenarios

Simulation + Control Tower Layer

OPERATIONAL NODES

45

ranked network

WAREHOUSES

20

inventory layer

DISTRIBUTION CENTERS

10

midstream layer

FACTORIES

5

production layer

SUPPLIERS

10

upstream layer

SCENARIOS

5

what-if engine

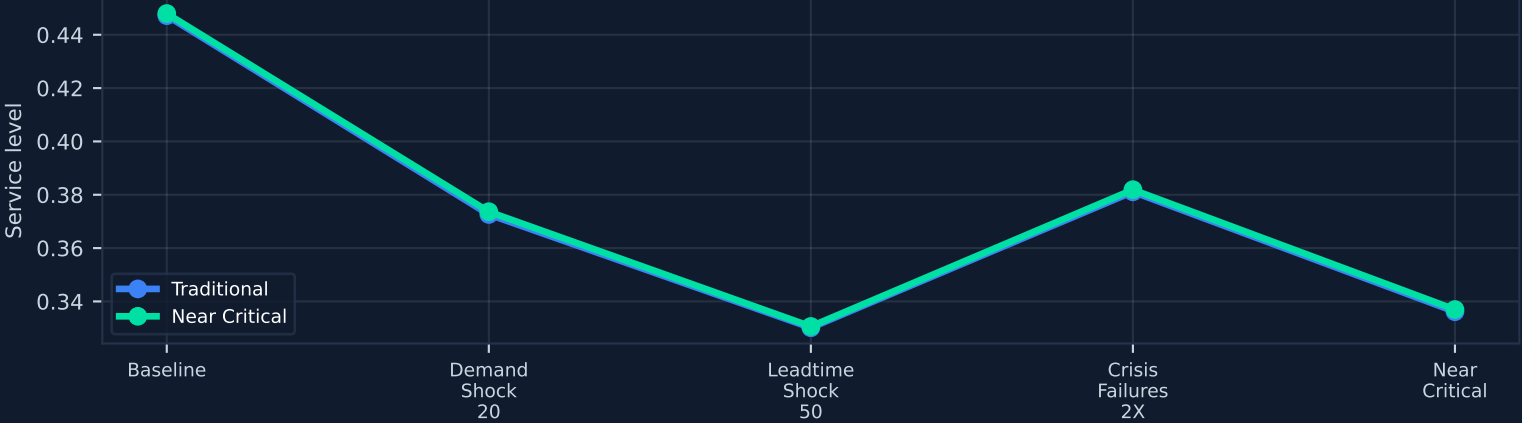
Synthetic Supply Chain Network

Command-center view of the multi-echelon system.



Scenario Stress Test

Service degradation under demand, failure and lead-time shocks.



Inventory & Backorder Dynamics

Digital twin state variables summarized by scenario.



Simulation Capabilities

- ✓ Multi-echelon network generation
- ✓ Dynamic inventory states
- ✓ Reorder points and target stock
- ✓ Stochastic lead-time shocks
- ✓ Demand and failure disruptions
- ✓ Monte Carlo scenario engine
- ✓ Automated output generation

Digital Twin Question

What happens if the system is stressed?

The platform evaluates shocks before they occur, then translates them into cost, service and intervention priorities.